

全等三角形常见模型

【模块一】一线三等角模型

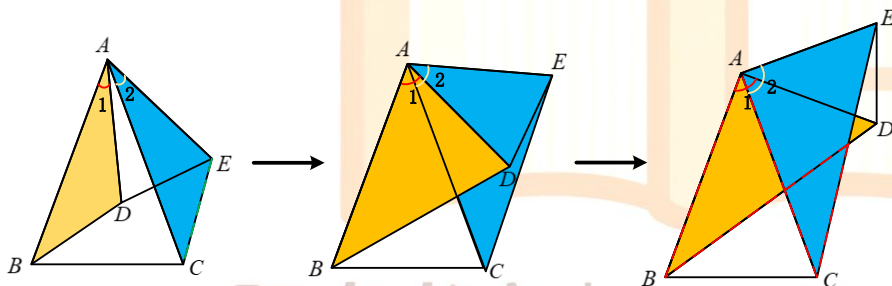
一线三垂直模型

全等图形				
已知条件	$\angle ABC = \angle DEF = \angle ACF = 90^\circ$ $AC = DF$	$\angle ABC = \angle DEF = 90^\circ$ $AC \perp DF$ $AC = DF$	$\angle ABC = \angle DEF = \angle ADF = 90^\circ$ $AC = DF$	
全等结论				

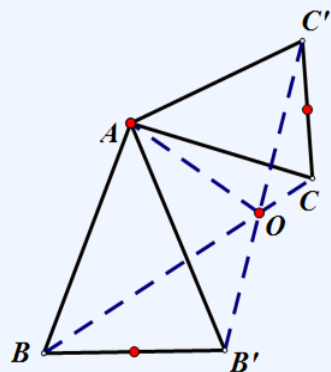
一线三等角模型

已知题千	(1) 角度关系: $\angle ADB = \angle ABC = \angle BEC$; (2) 边的关系: 三组对应边任意一组相等.	(1) 角的关系: 等边 $\triangle ABC$ 中, 将点 A 延 EF 折叠的对称点落到 BC 边上的点 D ; (2) 边的关系 $BE = CD$ 或 $BD = CF$.
对应模型		
结论		

【模块二】手拉手模型



如图, 已知 $\triangle ABB'$ 和 $\triangle ACC'$ 都是等腰三角形, $AB = AB'$, $AC = AC'$, 且 $\angle BAB' = \angle CAC'$.

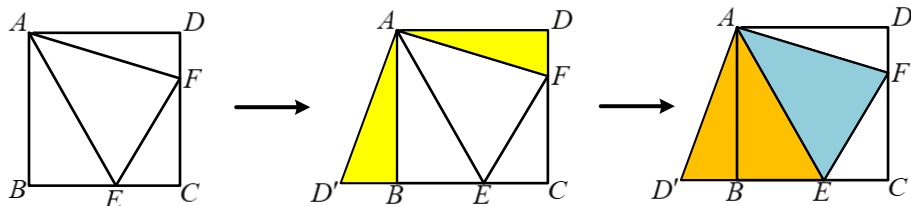


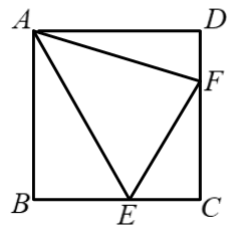
三个结论:

- (1)
- (2)
- (3)

【模块三】半角模型

1. 半角模型的一般情形



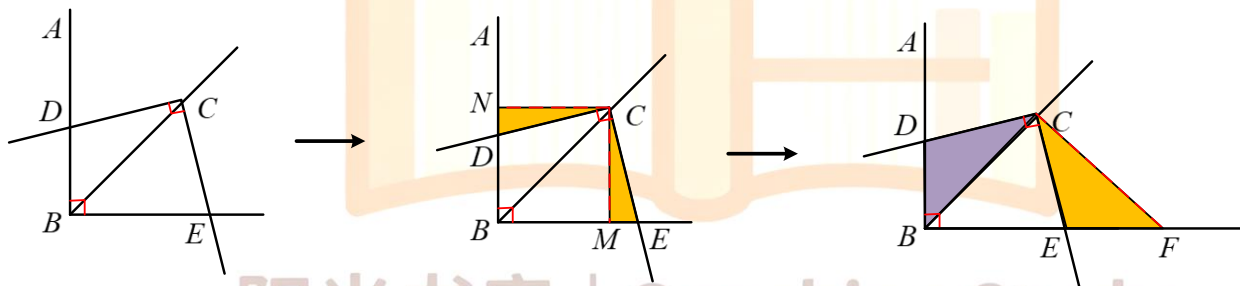
	(1) (2) (3)
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【模块四】对角互补模型

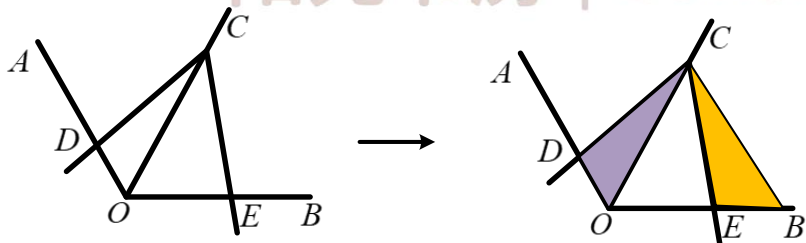
1. 全等类型---90°

【条件】(1) $\angle ABE = \angle CDE = 90^\circ$; (2) BC 平分 $\angle ABE$;

【结论】(1) $CD = CE$; (2) $BD + BE = \sqrt{2}BC$; (3) $S_{\text{四边形}DBEC} = S_{\triangle BCD} + S_{\triangle BCE} = \frac{1}{2}BC^2$.



2. 全等类型---120°



【条件】(1) $\angle AOB = 2\angle CDE = 120^\circ$; (2) OC 平分 $\angle AOB$;

【结论】(1) $CD = CE$; (2) $OD + OE = OC$; (3) $S_{\text{四边形}DOEC} = S_{\triangle OCD} + S_{\triangle OCE} = \frac{\sqrt{3}}{4}OC^2$.